THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$72 = 2^3 \times 3^2$; multiply by the primes with odd power
2	**(a)** 3^5	Add the exponents on top, then subtract the bottom exponent
3	**(d)** 22	Expand with powers of the base
4	**(c)** 75	$4x + 60 = 360$
5	**(c)** 2	Alternate digit sums must differ by 0 or 11
6	**(c)** 3	$192 = 2^6 \times 3$
7	**(d)** $\frac{9}{4}$	$(a/b)^{-n} = (b/a)^n$
8	**(c)** 101101	Repeated division by 2
9	**(c)** 20 cm	$2k(3+5) = 64$ gives $k = 4$
10	**(b)** 0	The number must be even and its digit sum a multiple of 3
11	**(d)** 9	Only the ones digit 7 matters: 7^2 ends in 9
12	**(b)** 7	$x + 4 = 11$
13	**(d)** CXI	44+67=111
14	**(b)** 34 cm	AO = OC; find x first, then AC = 2 AO
15	**(d)** 8262	Largest arrangement minus smallest arrangement

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$2916 = 2^2 \times 3^6$ (1); $\sqrt{2916} = 54$ (1).	1+1
17	$(2^3)^3 = 2^9$, so $2^9 \div 2^5 = 2^4$ (1) $= 16$ (1).	1+1
18	$1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$ (1); $= 53$ (1).	1+1
19	One part $= 360 \div 18 = 20^\circ$ (1); largest $= 160^\circ$ (1).	1+1
20	8456 divided by 11 leaves remainder 8 (1); add $11 - 8 = 3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$1872 = 2^4 \times 3^2 \times 13$ (1); multiplier 13 (1); product 24336, root 156 (1).	1+1+1
22	$2^2 \times 3^2$ (1+1); value $4 \times 9 = 36$ (1).	1+1+1
23	$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 45$ (1); $45 = 8 \times 5 + 5$; $5 = 8 \times 0 + 5$ (1); 55_8 (1).	1+1+1
24	$4x + 90 = 360$ (1); $x = 67$ (1); angles 67°, 82°, 97°, 112° (1).	1+1+1
	OR: LCM of 12 and 15 is 60 (1); first multiple not below 1000 (1); 1020 (1).	1+1+1
25	$12000 = 2^5 \times 3 \times 5^3$ (1); divisor 12 (1); quotient 1000, root 10 (1).	1+1+1
	OR: $\frac{1}{8} + \frac{1}{9}$ (1); common denominator (1); sum $\frac{17}{72}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Next perfect square above 1500 is $39^2 = 1521$ (2); students needed $= 21$ (2); rows $= 39$ (1).	2+2+1
	OR: Numerator power of 2: $2^3 \div 2^{-2} = 2^5$ (2); for 3: $3^{-1} \div 3^1 = 3^{-2} = \frac{1}{3^2}$ (1); result $\frac{2^5}{3^2}$ (1); value $\frac{32}{9}$ (1).	2+1+1+1
27	Add in binary: 11100_2 (3); decimal check $13 + 9 + 6 = 28$ (2).	3+2
	OR: Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $2x + 10 + 3x - 5 = 90$ gives $x = 17$ (2); $\angle A = 2 \times 44 = 88^\circ$ (1); $\angle B = 2 \times 46 = 92^\circ$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 1.2	1.44 is the square of 1.2
2	**(c)** 4	$3x = 12$
3	**(d)** 7	It is $111_2 = 2^3 - 1$
4	**(c)** 17	Adjacent angles are supplementary
5	**(c)** 60	Find the LCM
6	**(b)** 0.3	$0.027 = 0.3^3$
7	**(a)** 1000	$a^m \times b^m = (ab)^m = (10)^3$
8	**(c)** 5	20 lies between 2^4 and 2^5
9	**(a)** 85°	$\angle B = \angle D$ and the angle sum is 360°
10	**(a)** 3	4829 leaves remainder 5 when divided by 8
11	**(d)** 2025	Numbers ending in 5: 4×5 then 25 gives 2025
12	**(b)** 6×10^7	Multiply the numbers, add the exponents, then adjust
13	**(c)** 10001_2	11+6=17
14	**(a)** 85°	Exterior angles of a quadrilateral add up to 360°
15	**(d)** 1001	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 7

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{\sqrt[3]{27}}{\sqrt[3]{512}}$ (1) $= \frac{3}{8}$ (1).	1+1
17	$\frac{1}{32} = 2^{-5}$ (1); so $x = -5$ (1).	1+1
18	139 + 262 = 401 (1); CDI (1).	1+1
19	AO = OC gives $x = 4$ (1); AO $= 17$ cm so AC $= 34$ cm (1).	1+1
20	By 3: Yes, digit sum 15 is a multiple of 3 (1); by 4: Yes, last two digits 16 are divisible by 4 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Decimal values 5 and 3 (1); product 15 (1); 1111_2 (1).	1+1+1
22	Sides $= 2k$ and $5k$; $2(2k + 5k) = 56$ (1); $k = 4$ (1); sides 8 cm and 20 cm (1).	1+1+1
23	Alternate sums for each trial; need difference 0 or 11 (1); * $= 0$ (1); check 8305 (1).	1+1+1
24	$196 = 2^2 \times 7^2$ (1); root of the whole number 14 (1); so $\sqrt{1.96} = 1.4$ (1).	1+1+1
	OR: Total $= 2.5 \times 10^4 \times 12$ (1) $= 30 \times 10^4$ (1); in scientific notation 3×10^5 (1).	1+1+1
25	1994 = MCMXCIV, 449 = CDXLIX (1); difference = 1545 (1); MDXLV (1).	1+1+1
	OR: $\angle CAB = 90 - 25 = 65^\circ$ (1); OA = OB so $\angle AOB = 180 - 2(65) = 50^\circ$ (1); OC = OD so $\angle ODC = \angle OCD = \angle CAB = 65^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Steps: 9541-1459=8082; 8820-0288=8532; 8532-2358=6174 (3); total 3 steps (2).	3+2
	OR: $\sqrt[3]{1728} = 12$ cm (2); TSA $= 6s^2 = 6 \times 12^2 = 864$ cm² (3).	2+3
27	Number of cells $= \frac{8 \times 10^1}{2 \times 10^{-5}}$ (2); $= \frac{8}{2} \times 10^{1+5}$ (2); 4×10^6 (1).	2+2+1
	OR: Smallest $10000_2 = 16$ (2); largest $11111_2 = 31$ (2); count $32 - 16 = 16$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$98 = 2 \times 7^2$; multiply by the primes with odd power
2	**(d)** 431	$512 - 81 = 431$
3	**(c)** 100	Digit x (base)^(position from right, starting 0)
4	**(c)** rectangle	A parallelogram with equal diagonals
5	**(c)** 45	Subtract the reverse
6	**(a)** 2	$250 = 2 \times 5^3$
7	**(a)** $\frac{5}{6}$	$\frac{1}{2} + \frac{1}{3}$
8	**(b)** 27	Expand with powers of the base
9	**(b)** 60	$4x + 120 = 360$
10	**(c)** 2	Alternate digit sums must differ by 0 or 11
11	**(b)** 4	Only the ones digit 8 matters: 8^2 ends in 4
12	**(c)** 5^4	Add the exponents on top, then subtract the bottom exponent
13	**(b)** 110110	Repeated division by 2
14	**(a)** 18 cm	$2k(2 + 3) = 60$ gives $k = 6$
15	**(b)** 0	The number must be even and its digit sum a multiple of 3

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{\sqrt{81}}{\sqrt{256}}$ (1) $= \frac{9}{16}$ (1).	1+1
17	$3 \times 10^{8-3}$ (1) = 3×10^5 (1).	1+1
18	In decimal 26-9=17 (1); 10001_2 (1).	1+1
19	Fourth exterior $= 360 - 270 = 90^\circ$ (1); interior $= 180 - 90 = 90^\circ$ (1).	1+1
20	842-248=594 (1); $594 = 99 \times 6$, so divisible (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Middle number $= \frac{54}{3} = 18$ (1); numbers 17, 18, 19 (1); check sum $= 54$ (1).	1+1+1
22	$2646 = 2 \times 3^3 \times 7^2$ (1); multiplier 6 (1); product 15876, root 126 (1).	1+1+1
23	$3^2 \times 2^3$ (1+1); value $9 \times 8 = 72$ (1).	1+1+1
24	$1 \times 5^3 + 4 \times 5^2 + 3 \times 5^1 + 2 \times 5^0 = 242$ (1); $242 = 8 \times 30 + 2$; $30 = 8 \times 3 + 6$; $3 = 8 \times 0 + 3$ (1); 362_8 (1).	1+1+1
	OR: $4x + 48 = 360$ (1); $x = 78$ (1); angles 78°, 86°, 94°, 102° (1).	1+1+1
25	LCM of 14 and 21 is 42 (1); first multiple not below 1000 (1); 1008 (1).	1+1+1
	OR: $10976 = 2^5 \times 7^3$ (1); divisor 4 (1); quotient 2744, root 14 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Diagonals of a rectangle are equal: $3x + 5 = 2x + 15$ gives $x = 10$ (2); AC = BD $= 35$ cm (1); OA $= 17.5$ cm (1); wire $= 2 \times 35 = 70$ cm (1).	2+1+1+1
	OR: LCM(6, 8, 12) $= 24$ (3); required number $= 24 + 3 = 27$ (2).	3+2
27	Next perfect square above 2000 is $45^2 = 2025$ (2); students needed $= 25$ (2); rows $= 45$ (1).	2+2+1
	OR: Numerator power of 3: $3^2 \div 3^{-1} = 3^3$ (2); for 2: $2^{-1} \div 2^2 = 2^{-3} = \frac{1}{2^3}$ (1); result $\frac{3^3}{2^3}$ (1); value $\frac{27}{8}$ (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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